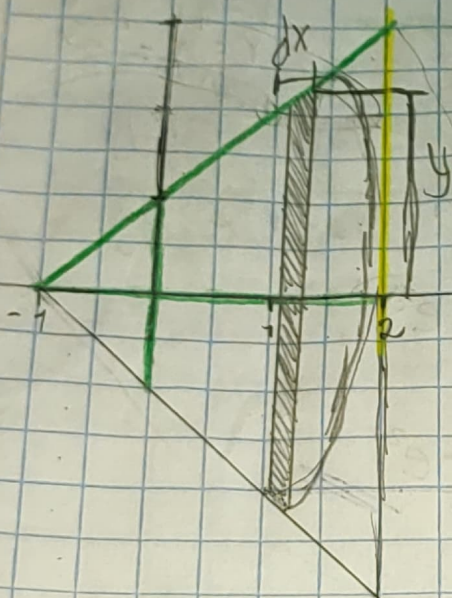


Tercera decisión 6.2 b

1) $y = x+1$, $y = 0$, $x = 0$, $x = 2$ @ eje X



$$V = \pi r^2 \quad r = y$$

$$V = \pi (x+1)^2 dx$$

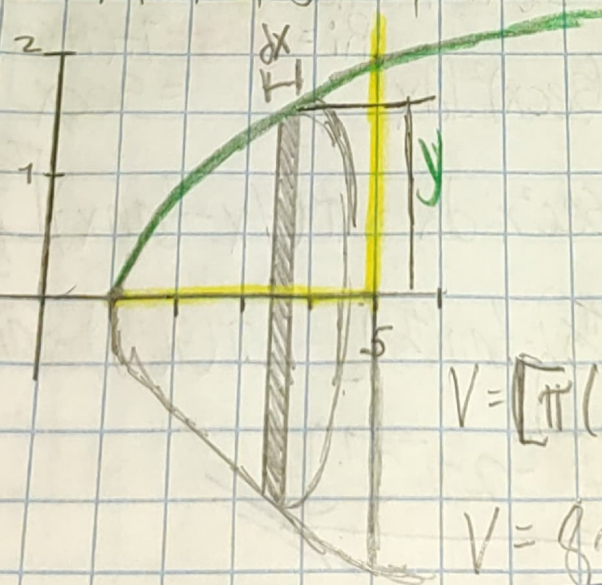
$$V = \pi \int_0^2 x^2 + 2x + 1 dx$$

$$V = \pi \left[\frac{1}{3}x^3 + x^2 + x \right]_0^2$$

$$V = \left[\pi \left(\frac{1}{3}(2)^3 + 2^2 + 2 \right) \right] - \left[\pi \left(\frac{1}{3}(0)^3 + 0^2 + 0 \right) \right]$$

$$V = \frac{26}{3} \pi$$

3) $y = \sqrt{x-1}$, $y = 0$, $x = 5$ @ eje X



$$V = \pi r^2 \quad r = y$$

$$V = \int_1^5 \pi (\sqrt{x-1})^2 dx$$

$$V = \pi \int_1^5 x-1 dx = \pi \left(\frac{1}{2}x^2 - x \right) \Big|_1^5$$

$$V = \left[\pi \left(\frac{1}{2}(5)^2 - 5 \right) \right] - \left[\pi \left(\frac{1}{2}(1)^2 - 1 \right) \right] = \frac{15}{2} \pi - \left(-\frac{1}{2} \pi \right)$$

$$V = 8 \pi$$

5) $y = \sqrt{25 - x^2}$, $y = 0$, $x = 2$, $x = 4$

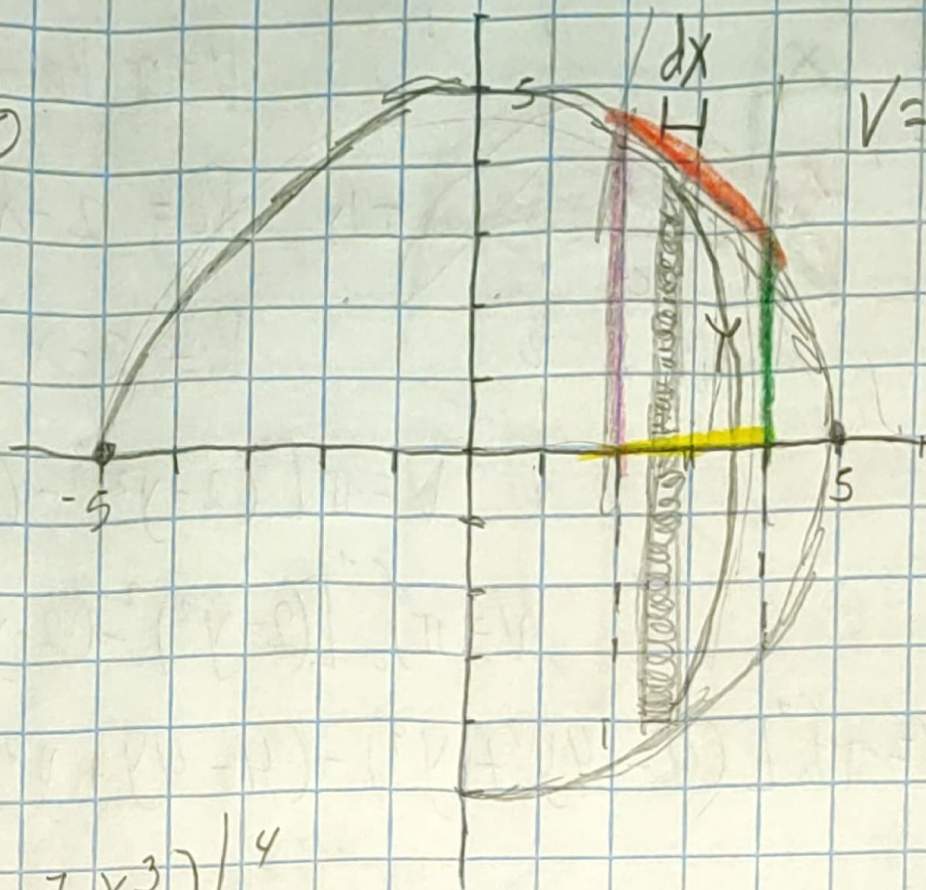
Intersecciones

$$\sqrt{25 - x^2} = 0$$

$$25 = x^2$$

$$y = 0, 5 = x$$

$$y = 0, -5 = x$$



$$V = \pi r^2 dx$$

$$V = \pi \int_2^4 (\sqrt{25 - x^2})^2 dx$$

$$V = \pi \int_2^4 (25 - x^2) dx$$

$$V = \pi \left(25x - \frac{7}{3}x^3 \right) \Big|_2^4$$

$$V = \pi \left[\left(25(4) - \frac{7}{3}(4)^3 \right) - \left(25(2) - \frac{7}{3}(2)^3 \right) \right] = \frac{236}{3}\pi - \frac{742}{3}\pi = \frac{94}{3}\pi$$

7) $y = \ln x$, $y = 1$, $y = 2$, $x = 0$

4) y

$$V = \pi r^2 dy$$

$$r = x = e^y$$

$$V = \pi \int_1^2 (e^y)^2 dy$$

$$V = \pi \int_1^2 e^{2y} dy$$

$$V = \pi \int_1^2 e^m \frac{dm}{2} = \frac{\pi}{2} \int_1^2 e^m = \frac{\pi}{2} \cdot e^{2y} \Big|_1^2$$

$$V = \left(\frac{\pi}{2} \cdot e^{2(2)} \right) - \left(\frac{\pi}{2} \cdot e^{2(1)} \right) = \frac{\pi}{2} \cdot e^4 - e^2$$

$$m = 2y$$

$$dm = 2dy$$

$$\frac{dm}{2} = dy$$

8ed-9) $y_1 = \frac{1}{4}x^2$

$y_2 = 5 - x^2$

ej. x

Intersecciones

$y_1 = y_2$

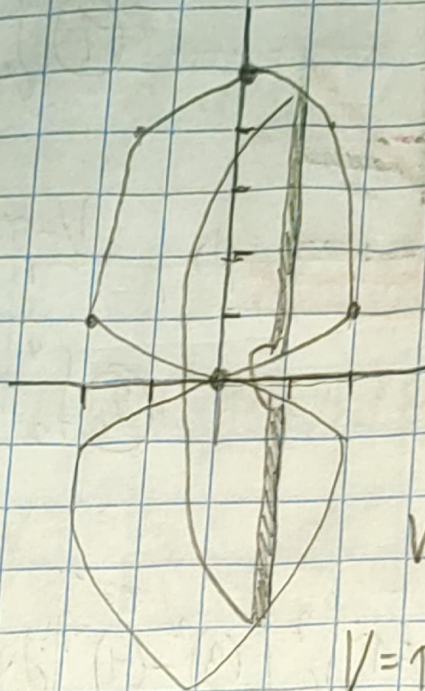
$\frac{1}{4}x^2 = 5 - x^2$

$\frac{5}{4}x^2 = 5$

$x^2 = 4$

$x = -2, y = 1$

$x = 2, y = 1$



$V = \pi(R_o^2 - R_i^2)dx$

$R_o = y = 5 - x^2$

$R_i = y = \frac{1}{4}x^2$

$V = \int_{-2}^2 \pi \left[(5 - x^2)^2 - \left(\frac{1}{4}x^2 \right)^2 \right] dx$

$V = \pi \int_{-2}^2 25 - 10x^2 + x^4 - \frac{1}{16}x^4 dx$

$V = \pi \left(25x - \frac{10}{3}x^3 + \frac{1}{16}x^5 \right) \Big|_{-2}^2$

$V = \frac{88}{3}\pi - \left(-\frac{88}{3}\pi \right) = \frac{176}{3}\pi$

11) $y = \frac{1}{4}x^2$, $x = 2$, $y = 0$ eje y

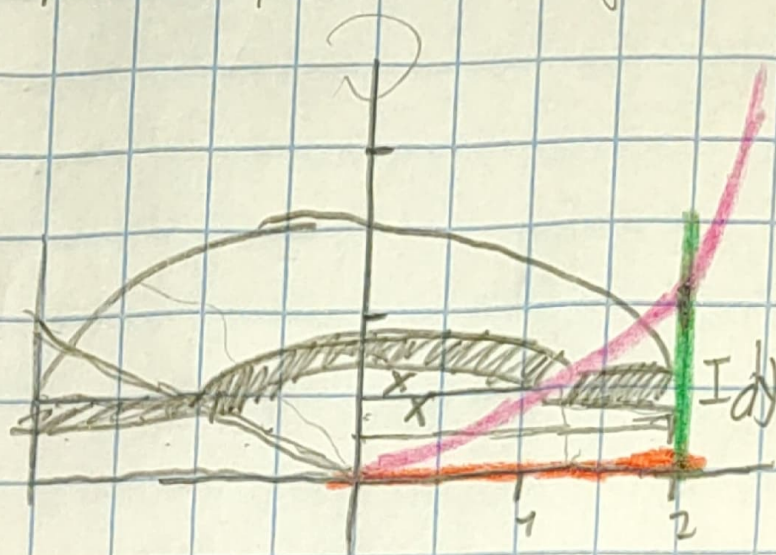
$$\frac{1}{4}x^2 = 0$$

$$x^2 = 0$$

$$x = 0$$

$$\sqrt{y} = \frac{1}{2}x$$

$$2\sqrt{y} = x$$



$$V = \pi [R_e^2 - R_i^2] dy$$

$$R_e = x = 2 \quad R_i = x = 2\sqrt{y}$$

$$V = \pi \int_0^1 [2^2 - (2\sqrt{y})^2] dy$$

$$V = \pi \int_0^1 4 - 4y dy = \pi (4y - 2y^2) \Big|_0^1$$

$$V = [\pi(4(1) - 2(1)^2)] = 2\pi$$

13) $y = 1 + \sec x$, $y = 3$

a) $y = 1$

$$V = \pi [R_e^2 - R_i^2] dx$$

$$V = \pi \int_{-\pi/3}^{\pi/3} [(2)^2 - (\sec x)^2] dx$$

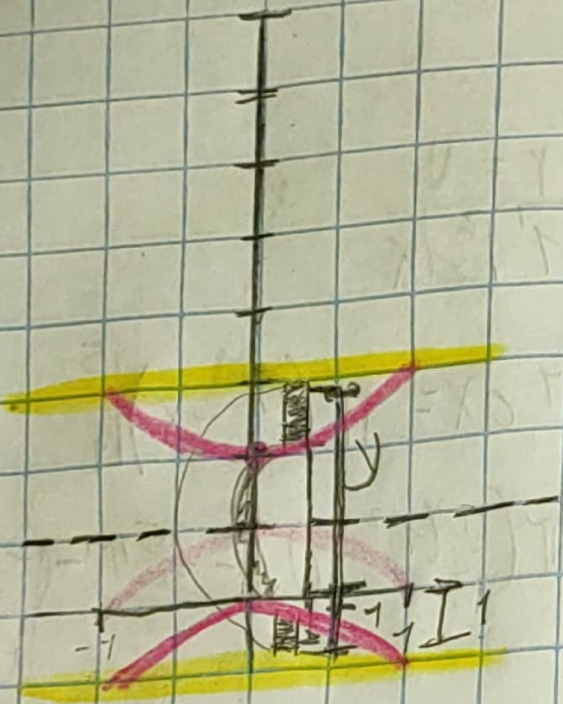
$$V = \pi \int_{-\pi/3}^{\pi/3} 4 - \sec^2 x \, dx = \pi (4x - \tan x) \Big|_{-\pi/3}^{\pi/3}$$

$$V = [\pi(4(\pi/3) - \tan(\pi/3))] - [\pi(4(-\pi/3) - \tan(-\pi/3))]$$

$$V = \pi \left[\frac{8\pi}{3} - 2\sqrt{3} \right]$$

$$R_e = y - 1 = 3 - 1 = 2$$

$$R_i = y - 1 = 1 + (\sec x) - 1 = \sec x$$

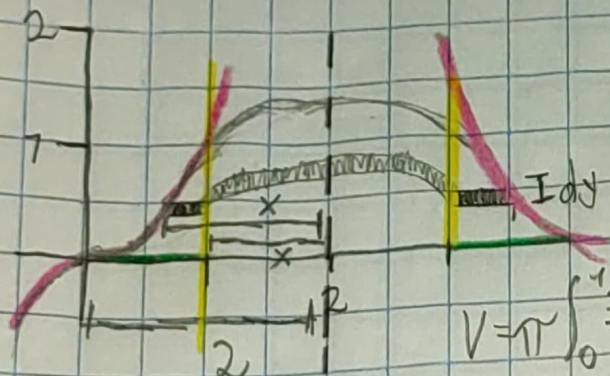


75) $y = x^3, y = 0, x = 1$

$x = 2$

$V = \pi [R_e^2 - R_i^2] dy$

$R_e = x = \sqrt[3]{y}$
 $R_i = x = 1$



$V = \pi \int_0^1 [(\sqrt[3]{y})^2 - (1)^2] dy$

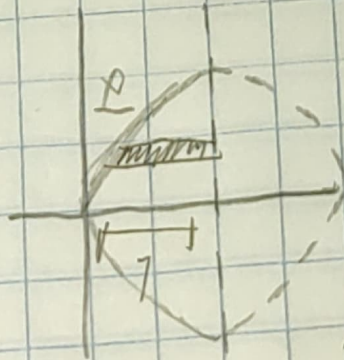
$V = \pi \int_0^1 (\sqrt[3]{y^2} - 1) dy = \pi \int_0^1 y^{2/3} - 1 dy$

$V = \pi \left(\frac{3}{5} y^{5/3} - y \right) \Big|_0^1 = -\frac{2}{5} \pi$ * Duda

$y = x^3$
 $x = \sqrt[3]{y}$

77) $x = x^2, x = 1, G(x) = 1$

$y^2 = 1 = y$
 $y_1 = 1$
 $y_2 = -1$ $x \neq 1$



$V = \pi r^2 = 1$

$V = \int_{-1}^1 \pi (1-x)^2 dy$

$\pi \int_{-1}^1 (1-2x+x^2) dx = \pi \left[x - \frac{2}{3} x^3 + \frac{1}{3} x^3 \right] \Big|_{-1}^1$

$= \frac{8}{15} + \frac{8}{15} = \frac{16}{15} \pi$